Physics Formulas and Key Points

Chap 1. Fundamentals of Kinematics

- **Relations between the basic kinematic quantities:**

$$\vec{r}(t) \Leftrightarrow \vec{v}(t) \Leftrightarrow \vec{a}(t)$$

(Note: the double arrows indicate that differentiation and integration are mutually inverse operations; pay attention to the initial conditions)

- **Normal (centripetal) acceleration:**

$$a_n = \frac{v^2}{R} = R\omega^2 \quad (v = R\omega)$$

- **Tangential acceleration:**

$$a_t = \frac{\mathrm{d}v}{\mathrm{d}t} = R\alpha$$

- **Matching textbook exercises:** 1-4, 1-6

Chap 2. Momentum and Energy

- **Cumulative effect of a force over time (impulse–momentum theorem):**

$$\vec{I} = \int_{t_1}^{t_2} \vec{F}\,\mathrm{d}t = \vec{p}_2 - \vec{p}_1$$

  - Matching textbook exercises: 2-7, 2-10

- **Cumulative effect of a force over space (work–energy theorem):**

$$W = \int_A^B \vec{F} \cdot \mathrm{d}\vec{r} = \Delta E_k$$

  - Matching textbook exercises: 2-16, 2-18

Chap 3. Rigid-Body Mechanics

- **Basic concepts:**
  - Moment of inertia I: thin rod, disk (pulley)
  - Torque:

$$\vec{M} = \vec{r} \times \vec{F}$$

  - Law of rotation:

$$\vec{M} = I\vec{\alpha}$$

- **Matching textbook exercises:** 3-6, 3-8, 3-10

Chap 5 + 6. Oscillations and Waves

- **Oscillation equation:**

$$x = A\cos(\omega t + \phi)$$

- **Wave equation:**

$$y = A\cos\left[\omega\left(t - \frac{x}{v}\right) + \phi\right]$$

- **Rotating vector (phasor) method:** Given the initial state $x_0, v_0 \implies \phi = -\frac{v_0}{\omega x_0}$ (see the physical meaning of the circular-motion projection for details)
- **Energy:**
  - Simple harmonic motion energy: E_k, E_p, E
  - Wave energy
- **Matching textbook exercises:** 6-5, 6-10

Chap 9. Electrostatic Field

- **Gauss's law and its applications:**

$$\oint \vec{E} \cdot \mathrm{d}\vec{S} = \frac{\sum q_{\text{enclosed}}}{\varepsilon_0}$$

(Applicable cases: spherical symmetry, axial symmetry, plane symmetry)

- **Electric potential:**

$$V_A = \int_A^{\infty} \vec{E} \cdot \mathrm{d}\vec{l} \quad (\text{spatial charge distribution})$$

$$V_A = \frac{Q}{4\pi\varepsilon_0 r_A} \quad (\text{point-charge potential})$$

- **Matching textbook exercises:** 9-9, 9-22, 9-12

Chap 10. Steady Magnetic Field

- **Ampère's circuital law:**

$$\oint \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 \sum I_i$$

- **Matching textbook exercises:** 10-9, 10-10, supplementary worked example